IPL Performance Analytics Dashboard

MINI PROJECT: IPL DATA ANALYSIS

Project Report

Prepared using Python, Pandas, NumPy, Matplotlib, Seaborn and Microsoft Power BI.

1. Project Objective

The objective of this project is to analyze IPL team and player performance, match outcomes, batting and bowling performance, venues, toss decisions and season-wise trends. The analysis converts IPL match data into meaningful statistical findings and an interactive Power BI dashboard.

2. Dataset Description

The project uses IPL match-level and ball-by-ball delivery data. Important fields include Match_ID, Season, Match_Date, City, Venue, Team1, Team2, Toss_Winner, Toss_Decision, Winner, winning margins, Player_of_Match, Innings, Batting_Team, Bowling_Team, Batsman, Bowler, Runs_Scored, Extra_Runs, Total_Runs, Wickets, Over and Ball.

3. Tools Used

Python; Pandas; NumPy; Matplotlib; Seaborn; Jupyter Notebook / Google Colab; Microsoft Power BI Desktop; Power Query; DAX.

4. Data Cleaning

The data was checked for data types, missing values and duplicate records. Duplicate rows were removed. Team-name inconsistencies caused by franchise renaming were standardized. Missing winners and player-of-match values were handled, dates were converted to datetime format, and super-over innings were excluded from the main delivery analysis. A cleaned ball-by-ball dataset and a cleaned match-level dataset were produced.

5. Data Analysis

The analysis covered total matches, matches by season, team wins, Player of the Match awards, venue usage, toss decisions, toss-result relationship, winning margins, leading run scorers, leading wicket takers, season-wise runs and wickets, average runs per match, team-wise average runs, season-wise team performance, venue performance and major IPL performance trends.

6. Python Visualizations

The Python notebook contains the following 14 visualizations required by the project brief:

1. Matches Played by Season — Bar Chart
2. Matches Won by Team — Bar Chart
3. Top 10 Run Scorers — Horizontal Bar Chart
4. Top 10 Wicket Takers — Horizontal Bar Chart
5. Player of the Match Awards — Bar Chart
6. Toss Decision Distribution — Pie Chart
7. Matches Won After Winning Toss — Bar Chart
8. Season-wise Total Runs — Line Chart
9. Season-wise Total Wickets — Line Chart
10. Runs Distribution — Histogram
11. Team Performance Comparison — Bar Chart
12. Venue-wise Match Count — Bar Chart
13. Runs vs Wickets — Scatter Plot
14. Correlation Heatmap

7. Power BI Dashboard

The completed dashboard contains the six required KPI cards, matches by season, team wins, top 10 batsmen, top 10 bowlers, runs by season, matches by venue, toss decision, Player of the Match, team performance matrix, and Season, Team and Venue slicers.

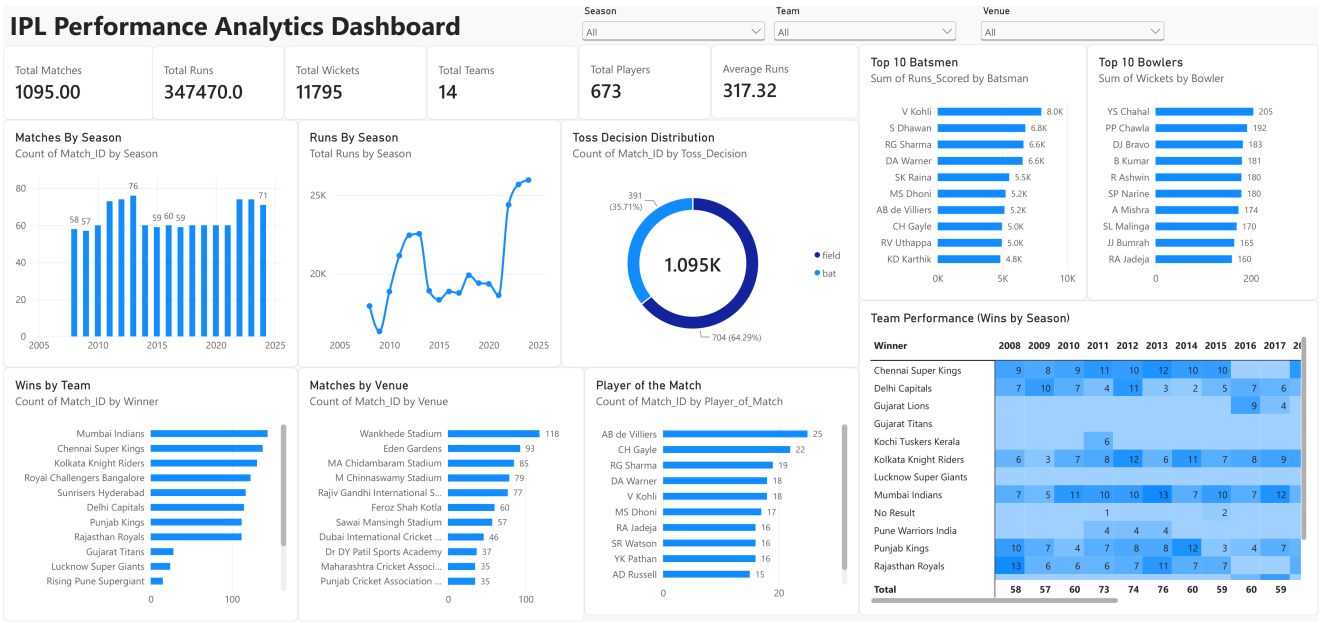


Figure 1. Completed IPL Performance Analytics Dashboard.

8. Business Insights

No.	Business Question	Finding
1	Which team has the highest number of wins?	Mumbai Indians
2	Which season had the highest number of matches?	2013, with 76 matches
3	Which player has received the highest Player of the Match awards?	AB de Villiers, with 25 awards
4	Who is the highest run scorer?	Virat Kohli (V Kohli), approximately 8.0K runs
5	Who is the highest wicket taker?	Yuzvendra Chahal (YS Chahal), with 205 wickets
6	Which venue hosted the highest number of matches?	Wankhede Stadium, with 118 matches
7	What is the most common toss decision?	Field, 704 matches (64.3%)
8	Does winning the toss appear to influence match results?	The toss winner won the match 50.83% of the time, indicating only a slight advantage.
9	Which season recorded the highest number of runs?	2024, with 25,971 runs
10	Which team has the highest average runs?	Gujarat Titans, with an average of 172.38 runs per innings
11	What are the major performance trends?	Average runs per match increased from 309.3 in the first season to 365.8 in 2024.

9. Key Analytical Insights

• Mumbai Indians recorded the highest number of wins in the analyzed dataset.
• The 2013 season recorded the highest number of matches, with 76 matches.
• AB de Villiers received the highest number of Player of the Match awards, with 25.
• Virat Kohli was the leading run scorer in the analysis, with approximately 8.0K runs.
• Yuzvendra Chahal was the leading wicket taker, with 205 wickets.
• Wankhede Stadium was the most frequently used venue, hosting 118 matches.
• Fielding was the most common toss decision, accounting for 64.3% of toss decisions.
• Winning the toss provided only a slight advantage because toss winners won 50.83% of matches.
• 2024 recorded the highest season total of 25,971 runs.
• Gujarat Titans had the highest average runs per innings at 172.38.
• Average runs per match increased from 309.3 in the first season to 365.8 in 2024.

10. Conclusion

The IPL data analysis provides a consolidated view of team performance, player contributions, toss behavior, venues and season-wise scoring patterns. The Python analysis was used for data cleaning, statistical analysis and visualization, while Power BI was used to create an interactive dashboard. The findings show strong individual and team performances, a preference for fielding after winning the toss, and an overall increase in scoring rates across the analyzed seasons.